Two-stage VLB + ELB Batch

Build and go-live report | Caliber Pulse-II MIS | branch feature/vlb-elb-two-stage | 01 November 2026

1. What this delivers

One batch per site, date and shift now carries both the tipper cards (Vehicle Log Book, VLB) and the machine cards (Equipment Log Book, ELB). The batch locks its tipper side first, then walks the operator through every machine booked at the site, and closes only when every machine has a saved shift status. A loading machine from another site holds its cards until the asset is transferred, and the transfer can be completed from the entry screen with one OTP read out by the Project Manager of the site that currently holds the asset. A brand-new asset can be created the same way with an OTP to the entering site's PM.

The two-stage flow is on for all users by default. A server switch, VLB_TWO_STAGE_BATCHES=0, returns the site to the single-step Save & lock batch without a redeploy, which is the rollback. Decision of 1 November 2026: go live for all users directly rather than through a separate test platform.

2. Decisions this build follows

Topic	Decision applied
Trip cross-check	None. VLB trip rows are the single truth for trips per machine. The ELB card asks no trip count.
Machine identity	Equipment master id. V19, VL19 and VPC 019 reconcile to one machine.
ELB stage scope	The whole site fleet from the Pulse-synced Equipment master, not only loaders. Dozers, graders and machines in maintenance all appear.
Zero trips	Allowed. Working needs working hours greater than zero; zero trips and zero hours must be maintenance, no operator, breakdown or not in use.
Asset from another site	Card is held, never rejected, and no OTP is asked for VLB or ELB entry. One OTP to the PM of the asset's current site completes an express transfer.
Asset not in any master	Created from the entry screen with chassis or registration mandatory and an OTP to the entering site's PM.
Equipment master authority	Pulse only. Pulse-II reads it through the existing one-way sync. Express writes are recorded local-only until Pulse exposes write endpoints.
Open batches	One open batch per site is kept, because three shifts run per day.
Rollout	Live for all users on pulse-ll.cmll.in from the first deployment; VLB_TWO_STAGE_BATCHES=0 is the rollback switch.

3. Batch lifecycle

Stage	What the operator can do	How it moves on
VLB open	Add, correct or delete tipper cards by manual entry, scan or PDF upload. ELB entry is disabled for this site, date and shift.	Lock VLB. Blocked while any card is awaiting a transfer, any loading machine does not resolve to one master record, or the batch is empty.
ELB open	Tipper cards are frozen. The fleet list opens with trips pre-filled. Work through machines in the stepper, or scan and upload ELB cards.	Close batch. Blocked until every machine in the fleet list has a saved status and no card is awaiting a transfer.
Closed	Reports and exports.	Reopen through the existing PM approval; the batch returns to ELB open. A site PM or MIS administrator can also Unlock VLB with a reason.

At Lock VLB the server stores two frozen lists with the batch: the machine snapshot (every loader named on a tipper card with trips and source cards) and the fleet list (every equipment-kind asset booked at the site at that moment). A later master change cannot alter what the shift was asked to account for.

4. The machine stepper

In the ELB stage the operator presses Start machines. A dialog opens on the first machine without a saved status, loaders first by trips, then the rest by door number. The header shows the machine, trips loaded, the tipper cards that named it, and progress such as 7 of 23 saved.

Shift status	Required	Trips allowed
Working	operator, opening and closing HMR, working hours above zero	any
Working, partial breakdown	operator, HMR, breakdown hours below the shift, reason	any
Breakdown	reason; breakdown hours set to the full shift	0
Maintenance	reason or job card number	0
No operator	nothing	0
Not in use	nothing	0

- A machine with trips cannot be saved as breakdown, maintenance, no operator or not in use. The dialog explains: “VPC-019 loaded 42 trips on 5 tipper cards this shift.”
- Next saves the machine as a real equipment log book linked to the batch and opens the next unsaved one. Previous goes back. Skip leaves the machine for later; skipped machines block closure by name.
- Every field change auto-saves a draft after a short pause. Reopening the batch resumes at the first incomplete machine.
- ELB cards saved through the normal form, scan or upload join the same batch and fill that machine’s row. While VLB is still open they are refused with “Lock VLB first”.

5. Assets from another site and new assets

Hold until transferred

When a tipper door, a loading machine on a trip row, or an ELB machine belongs to another site in the Equipment master, the card is saved with a hold instead of being rejected. The hold is not a stored status: on every batch read, Lock VLB and Close batch the server re-checks the asset against the current master. The moment the master shows the asset at this site, every card naming it is a normal card. Lock and close messages list holds per asset, for example: “Awaiting transfer: VPC-077 from Jayant OC (1 card, 4 trips)”.

Express transfer by OTP

- The operator presses Express transfer (OTP) on the asset row. The server creates a record in the existing Vehicle transfers workflow, source = asset’s current site, destination = batch site, or reuses a pending transfer to the same destination.
- One message goes to every PM of the source site, Telegram first, WhatsApp fallback through the approved authentication template. If neither is registered the request is refused and the normal workflow remains.
- The PM reads the code to the operator. On a correct code the three workflow steps complete in one transaction: source approval recorded as the PM(s) who received the OTP, destination verification as the requesting MIS user, destination acceptance recorded as implied. The Equipment master copy updates, destination PMs get a bell notification.
- Controls: six random digits, hash only, single use, ten minutes, five wrong attempts, one request per minute. These match the Vehicle transfers OTP note in the docs folder.

New equipment by OTP

- A door that matches nothing in the master offers Create new equipment: door, category (Vehicle or Equipment), name, make, model, registration, chassis; site fixed to the batch site. Chassis or registration is

mandatory because the Pulse sync links on them.

- The server first searches every site by chassis, then registration, then door. A match means a transfer, not a new asset, and the response says so.
- One OTP goes to the entering site's PM. On success the asset is created with its site, who approved it and who entered it, and the waiting card can be saved.
- Only available from an open batch; it is not a general master shortcut.

Pulse as the only master

Pulse-II still reads the Equipment master from Pulse. Until Pulse offers two write endpoints, complete a transfer and create an equipment record, each express action updates only the Pulse-II synced copy and stamps pulseWrite.mode = local-only on the transfer or asset so the Pulse administrator can follow up. The sync preserves these local records because a transferred record keeps its transfer date and an unlinked record is matched by chassis, door or registration.

6. Machine-wise shift report

A closed or ELB-open batch shows a second table under the batch report. Loaders first: door, equipment, trips, tipper cards, operator, opening and closing HMR, hours, trips per hour, breakdown hours, status, ELB number. Then Other equipment: door, equipment, status, hours, idle, remark. A totals row and a fleet summary line: machines, working, down, idle.

7. Screens changed

- Create Batch: a step indicator 1 VLB · 2 ELB · Closed; Lock VLB with blocker text; Start / Continue machines; ELB manual / scan; Create new equipment; Unlock VLB for PM or administrator; Close batch replaces Save & lock batch when the flag is on; batch list shows VLB open / ELB open / Closed.
- Fleet table with door, equipment, trips, status or Not saved, and entry column (ELB number, Skipped, Draft).
- Awaiting transfer panel with per-asset rows, Express transfer (OTP), OTP input, countdown and resend.
- Batch report with the machine-wise table and totals.
- New components: src/vlb-fleet-stepper.jsx, src/vlb-batch-holds.jsx. Edited: src/vlb-create-batch.jsx, src/vlb-batch-report.jsx, src/vlb-create-batch.css, one line in src/main.jsx.

8. Server changes

File	Purpose
vlb-two-stage.mjs (new)	Feature flag, additive schema, and the pure rules: machine snapshot, fleet list, shift status validation, lock and closure blockers, hold evaluation, machine report rows.
vlb-batch-stages.mjs (new)	Lock VLB, Unlock VLB, fleet rows, draft auto-save, stepper save as an equipment log book, normal ELB saves joining the batch, closure assertion, stage details for the batch API.
otp-approval.mjs, otp-delivery.mjs (new)	Shared OTP controls and delivery: Telegram outbox first, WhatsApp fallback to the registered mobile.
vehicle-transfer-express.mjs (new)	Express transfer request and verify routes; completes the three workflow steps in one transaction; pending transfer lookup for the holds panel.
equipment-create-express.mjs (new)	New equipment request and verify routes with cross-site duplicate detection.
vlb-batches.mjs	Stage details in every batch response, new routes lock-vlb, unlock-vlb, fleet, fleet draft, fleet save; close runs the two-stage checks; reopen restores the right stage.
log-book.mjs	Holds instead of rejection for other-site assets when the flag is on; shift status and reason on equipment books; operator and HMR optional for non-working statuses; ELB linking after save.

trip-equipment-validation.mjs	Optional any-site resolution so a loader from another site resolves and carries its site for the hold.
vlb-batch-policy.mjs	Two-stage schema appended; tipper cards frozen once VLB is locked.
server.mjs	Registers the two OTP modules with the real access checks, PM lookup, Telegram and WhatsApp delivery, the local-only Pulse writer and audit events.

New tables (all additive)

```
mis_vlb_batches: stage, vlb_locked_at, vlb_locked_by
mis_vlb_batch_machines, mis_vlb_batch_fleet, mis_elb_batch_entries
mis_vehicle_transfer_otps, mis_equipment_create_requests
```

New routes

```
POST /api/vlb-batches/:id/lock-vlb
POST /api/vlb-batches/:id/unlock-vlb {reason}
GET  /api/vlb-batches/:id/fleet
PUT  /api/vlb-batches/:id/fleet/:assetId/draft {draft}
POST /api/vlb-batches/:id/fleet/:assetId {shiftStatus, employeeId, HMR, ...}
POST /api/vehicle-transfers/express {batchId, assetId}
POST /api/vehicle-transfers/express/:otpId/verify {otp}
POST /api/equipment/express {batchId, draft}
POST /api/equipment/express/:id/verify {otp}
```

9. Verification

Check	Result
New backend tests (test/vlb-two-stage.test.mjs, test/express-otp.test.mjs)	9 tests pass, including a full API run: create, held card, lock refused, transfer clears the hold, lock, stepper rules, closure refused and then allowed.
New UI tests (test/vlb-two-stage-ui.test.mjs)	4 tests pass; existing batch UI tests unchanged and passing.
Full suite on this laptop	3137 pass, 4 fail, 19 skipped.
Production build (vite build)	Succeeds.

The four failures are environment issues on this Windows machine and are unrelated to the module: a symlink permission error in the release packaging test, a missing Python interpreter for the Azure Gemini configuration helper test, a locked temp folder in the startup gate test, and a pre-existing line-ending bug in the batch deletion UI test on Windows checkouts. None of the files they exercise were changed. Gemini itself is configured and working on the live server; that test only exercises a one-time deployment helper.

Not yet done: the screens have not been clicked through against a live database. That is the first step on the test platform.

10. Switches and rollback

Switch	Live value	Effect
VLB_TWO_STAGE_BATCHES	unset (on)	Set to 0 on the App Service to return to single-step batches at once; new tables stay but unused.
Pulse write endpoints	not available yet	Express transfers and new assets update the Pulse-II synced copy and record local-only; follow up in Pulse.
OTP delivery	MIS Telegram bot linked chats, WhatsApp fallback	PMs without a linked Telegram or registered mobile approve through the normal transfer workflow.

- Deployment: merge to main; GitHub Actions runs the tests on Linux and deploys to App Service caliber-mis-783 (pulse-ll.cmll.in). Browsers reload on the new app version.

- First shift after go-live: open batches created before the deployment start in VLB open; lock VLB, then enter machines. ELB cards for a site, date and shift whose batch is still VLB open are refused until Lock VLB.
- Watch for: PMs without Telegram or mobile registered (express OTP refused), machines booked at the wrong site in Pulse (they appear in the wrong fleet list), and local-only Pulse writes needing follow-up.
- Deploy the branch to a separate URL with its own database restored from a live backup, Pulse masters read-only, WhatsApp and email disabled, TEST banner on.
- Sites run real shifts on both systems for an agreed period. Check time per shift in the stepper, how often transfers and new-asset OTPs are needed, and whether closure messages are clear.
- Build the team’s change list, re-test, then switch the flag on for one live site, then all. Switching the flag off is the rollback.
- Open items before go-live: the two Pulse write endpoints, a per-site option for a second OTP from the destination PM, and the team’s change list.